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Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN12601)

Lab Manual: Assignment-2

Lab Title: Programs on Lists, Tuple, and Dictionary in Python

Objective:

1. To understand and practice basic operations on Python data structures: lists, tuples, and dictionaries.
2. To learn how to manipulate lists using various methods such as insert, remove, append, and clear.
3. To explore tuple operations including adding items, accessing elements, and checking for item existence.
4. To implement dictionary operations, including storing and retrieving key-value pairs.
5. To solve problems involving comparisons of data structures and identifying duplicate elements in lists.

Materials Required:

- Computer with internet access
- Python (latest version)
- Python Integrated Development Environment (IDLE)
- Access to an online Python environment (e.g., Google Colaboratory)

Theory:

Lists, tuples, and dictionaries are fundamental data structures in Python, each serving different purposes:

Lists are mutable sequences that allow you to store multiple items in a single variable. They support a variety of operations such as insertion, deletion, and concatenation, making them versatile for managing collections of items. Common list methods include:

- insert(): Adds an element at a specified position.
- remove(): Removes the first occurrence of a specified element.
- append(): Adds an element at the end of the list.
- pop(): Removes and returns the last item (or the item at a given index).
- clear(): Removes all items from the list.

Tuples are immutable sequences that are used to store multiple items in a single variable, similar to lists, but they cannot be altered once created. This immutability makes tuples useful for storing data that should not change throughout the program's execution. Tuples support operations like accessing items, checking for membership, and determining their length.

Dictionaries are collections of key-value pairs. They are unordered, mutable, and indexed, providing a flexible way to store and retrieve data efficiently. Common operations include adding, updating, and deleting key-value pairs, making dictionaries suitable for representing structured data, such as a set of names and corresponding marks.

List of Programs in Assignment-2

1) Create a list and perform the following methods
    a) insert()
    b) remove()
    c) append()
    d) len()
    e) pop()
    f) clear().
2) Write a Python function that takes two lists and returns True if they are equal otherwise false.
3) Create a tuple and perform the following methods
    a) Add items
    b) len()
    c) check for item in tuple
    d) Access items.
4) Write a Python program to enter name and percentage marks in a dictionary and display information on the screen.
5) Write a program print duplicate element present in the list